

REVENUE DECLINE INVESTIGATION

Why Revenue declined by 43%?

This project investigates the decline across two areas: Product and Geography. A drill-down approach is used to identify the root causes and understand the key contributors to the decline. Additionally, Financial Impact and Recovery Analysis were conducted to assess the overall business performance.

The goal is to identify the contributing factors, understand the financial impact of the decline, and pinpoint areas of recovery, so the business can understand what led to the decline.

Overview

- 01 Executive Overview
- 02 Root-cause Analysis
- 03 Financial Impact
- 04 Recovery Analysis
- 05 Recommendation
- 06 Additional Information (ERD | Project Workflow | Analytical Workflow)

About this Project

TOOLS SQL | Power BI | MS-Excel

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Executive Overview

Introduction

Executive Overview

Root-cause Analysis

Financial Impact

Recovery Analysis

Recommendations

Increase

Decrease

Neutral

Select Category

All

Select Subcategory

All

Select Continent

All

Select Country

All

Clear all slicers

Revenue %YoY

-43.1% ▼

Revenue 2011

\$152.2M

Revenue 2012

\$86.6M

Revenue Change

-\$65.6M

Gross Profit %YoY

-43.7% ▼

Gross Profit 2011

\$89.3M

Gross Profit 2012

\$50.3M

Gross Profit Change

-\$39.0M

Quantity Sold %YoY

-41.9% ▼

Quantity Sold 2011

442,660

Quantity Sold 2012

257,330

Quantity Sold Change

-185,330

Avg. Selling Price %YoY

-2.2% ▼

ASP 2011

\$343.8

ASP 2012

\$336.4

ASP Change

-\$7.4

Revenue Trend Analysis

2011 2012

Revenue

Gross Profit

Quantity Sold

ASP

0.5M

Apr

May

Jun

Jul

Aug

Sep

Oct

0.0M



Apr, May, June, July and October are selected for further investigation based on their contribution to the Revenue and Quantity Sold decline.

Revenue Recovery

Apr

-46.3%

May

-47.6%

Jun

-49.2%

Jul

-43.5%

Aug

5.6%

Sep

3.4%

Oct

-40.2%

-40%

-20%

0%

Monthly Revenue Contribution %YoY

Month

%Revenue

%Gross Profit

%Quantity Sold

%Contribution by Month

Apr

-46.3%

-47.1%

-42.7%

-19.9%

May

-47.6%

-48.1%

-47.0%

-27.6%

Jun

-49.2%

-49.7%

-48.0%

-27.5%

Jul

-43.5%

-44.5%

-42.8%

-4.3%

Aug

5.6%

5.4%

15.8%

0.9%

Sep

3.4%

1.9%

19.2%

1.1%

Oct

-40.2%

-40.4%

-41.9%

-22.8%

Total

-43.1%

-43.7%

-41.9%

-100.0%

Root-Cause Analysis

Quantity Sold Change
-185K

Bottom 1 Class
103%
% of Decline

Bottom 5 Categories
98.8%
% of Decline

Bottom 9 Subcategories
86.7%
% of Decline

Bottom 4 Manufacturers
79%
% of Decline

Product

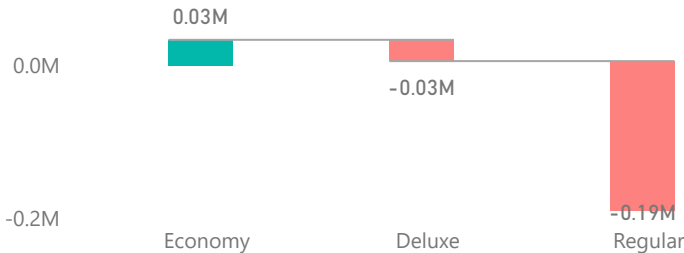
Geography

Each chart below drills into the previous one: Class to Category, Category to Subcategory, and Subcategory to Manufacturer, using the bottom performing segments at each level.

Quantity Change by Class

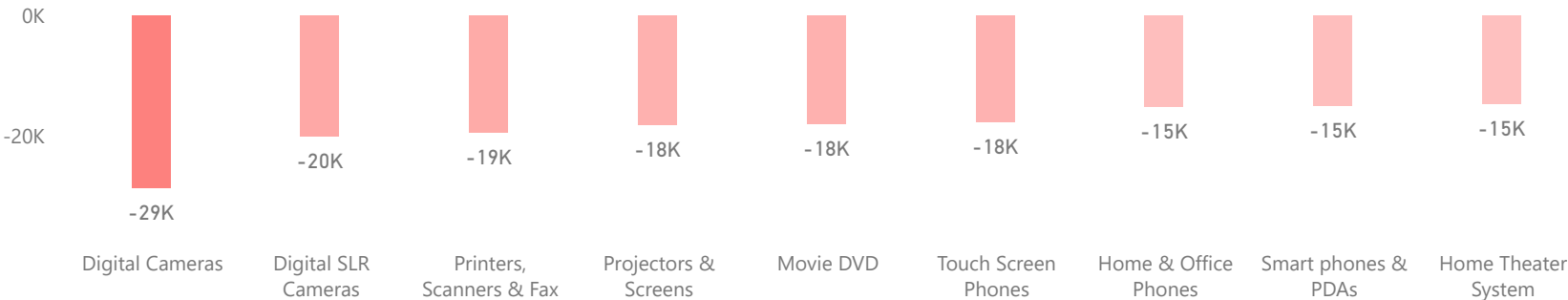
(within Apr, May, Jun, Jul & Oct)

IncreaseDecrease



Quantity Change of Bottom 9 Subcategories

(within previous 5 Categories)



Category Performance by Quantity Change

(within Regular Class)

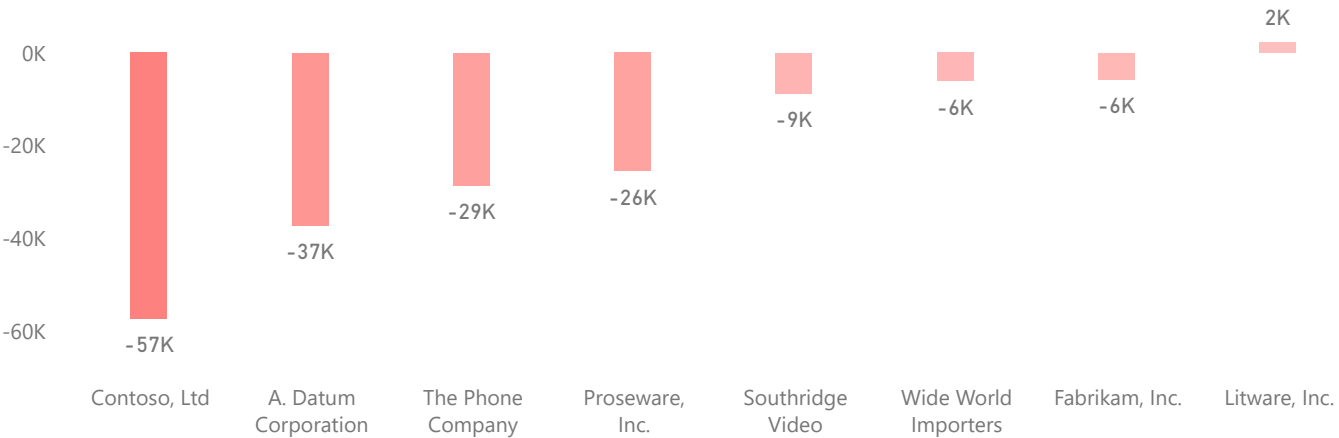
Details

Visual

Product_Category	Quantity Change	%YoY	%Contribution by Category
Audio	-4,340	-58.6%	-2.3%
Cameras & Camcorders	-55,730	-51.0%	-30.1%
Cell Phones	-51,500	-51.1%	-27.8%
Computers	-31,120	-24.9%	-16.8%
Music, Movies & Audio Books	-23,020	-52.8%	-12.4%
TV & Video	-19,620	-34.6%	-10.6%
Total	-185,330	-41.9%	-100.0%

Quantity Change by Manufacturer

(within previous 9 Subcategories)



Financial Impact

Revenue %YoY-43.1% ▼



Gross Profit %YoY-43.7% ▼



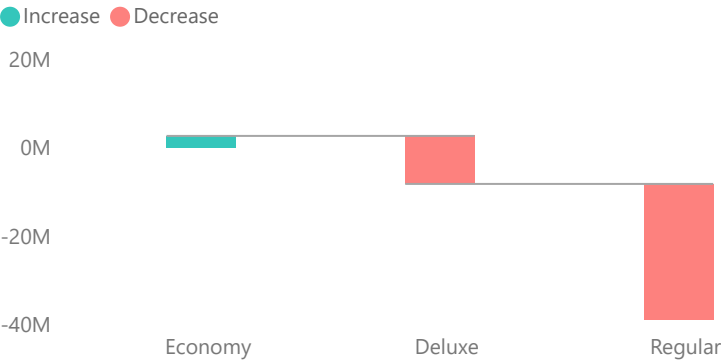
Cost %YoY-42.4% ▼



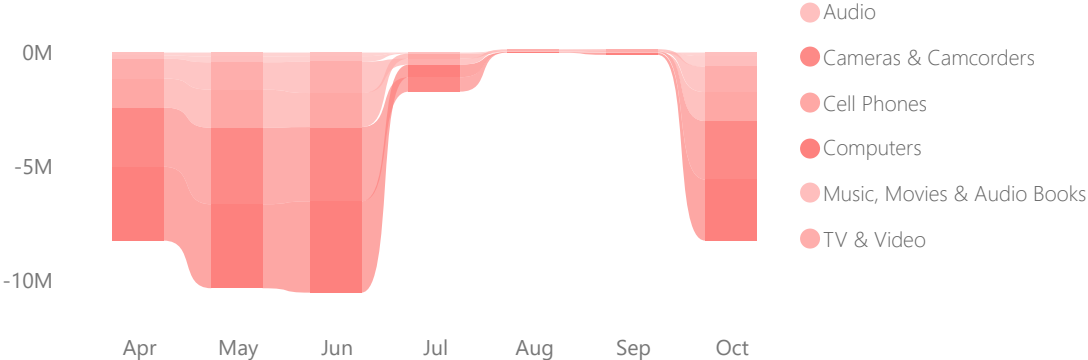
Profit Margin %YoY-42.6% ▼



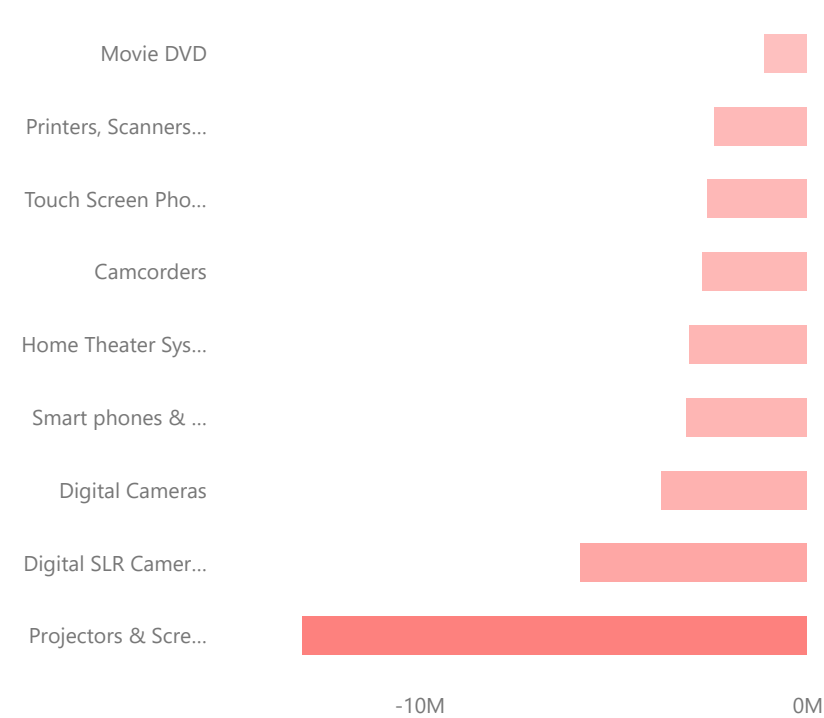
Profit Change by Product Class



Profit Change Trend by Product Category



Profit Change by Bottom 9 Product Subcategories



Profit Change by Country



Recovery Analysis

Product

Geography & Store



Each chart below drills into the previous one: Class to Category, Category to Subcategory and Subcategory to Manufacturer, using the top performing segment at each level.

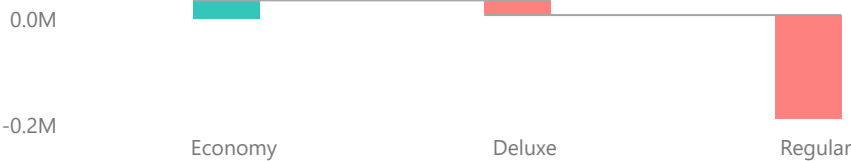
Revenue %YoY -79.9% ▼

Revenue 2011	\$148.1M
Revenue 2012	\$82.3M
Revenue Change	-\$65.8M

Quantity Change by Product Class

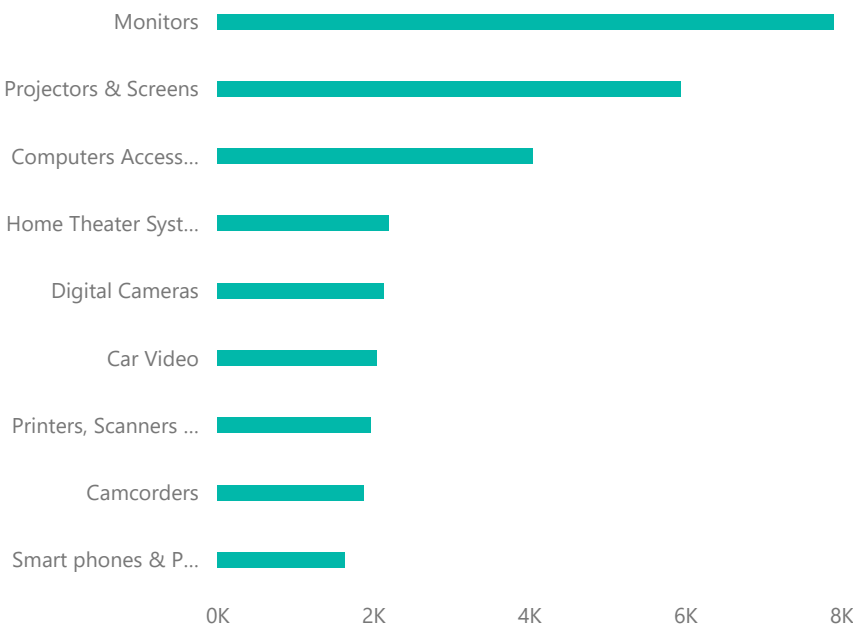
(Apr, May, Jun, Jul & Oct)

● Increase ● Decrease



Quantity Change by Top 9 Subcategories

(within previous 4 Categories)

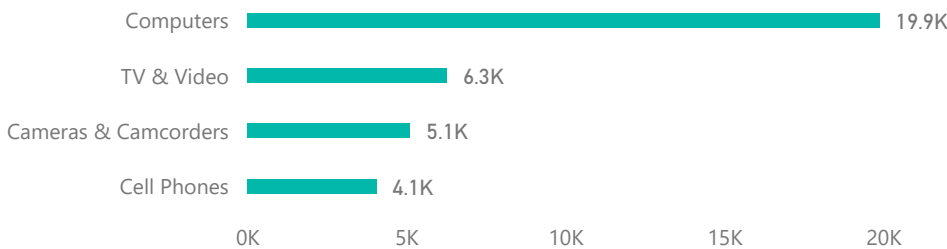


Quantity Sold %YoY -43.4% ▼

Quantity Sold 2011	431,750
Quantity Sold 2012	244,320
Quantity Sold	-187,430

Quantity Sold Recovery by Category

(within Economy)

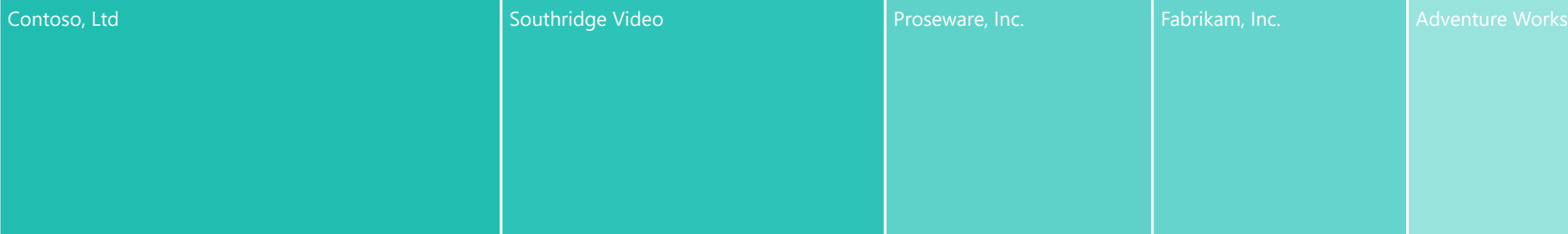


Avg. Selling Price %YoY -4.4% ▼

ASP 2011	\$334.6
ASP 2012	\$319.9
ASP Change	(\$14.6)

Quantity Change by Top 5 Manufacturers

(within previous 9 Subcategories)



Recommendations

IntroductionExecutive OverviewRoot-cause AnalysisFinancial ImpactRecovery AnalysisRecommendations

CriticalHighLow

i Limited available data. Some recommendations require further investigation & validation before action

Revenue %YoY -43.1% ▼

Revenue 2011 \$152.2M
Revenue 2012 \$86.6M
Revenue Change -\$65.6M

Gross Profit %YoY -43.7% ▼

Profit 2011 \$89.3M
Profit 2012 \$50.3M
Profit Change -\$39.0M

Quantity Sold %YoY -41.9% ▼

Quantity 2011 442,660
Quantity 2012 257,330
Quantity Change -185,330

ROOT-CAUSE ANALYSIS

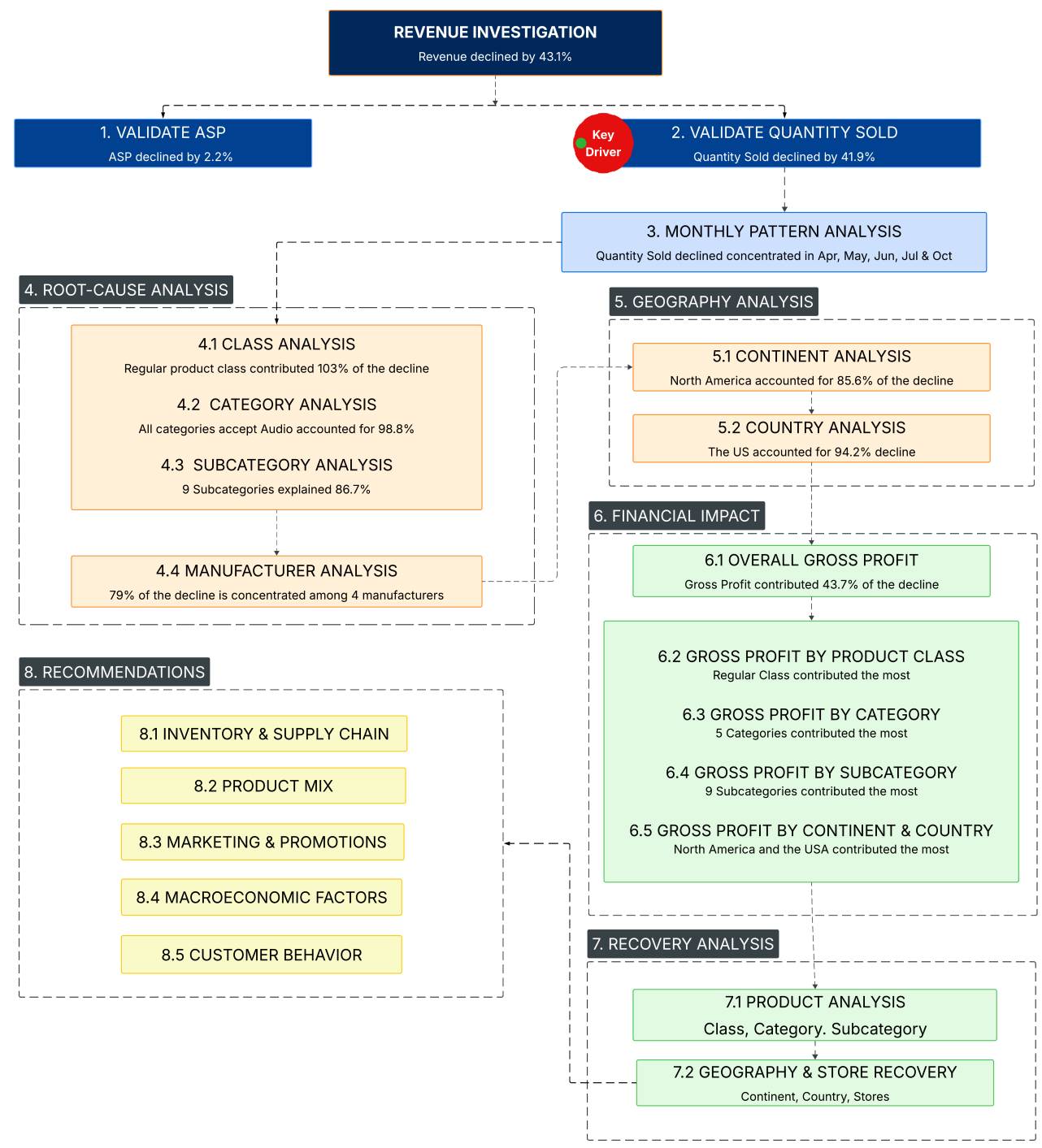
Key Drivers	Period	Product	Manufacturer	Geography
41.9% decline in Quantity Sold contributed to the 43.1% revenue decline.	185K unit decline concentrated in Apr, May, Jun, Jul & Oct.	103% of the quantity decline came from Regular class. 98.8% from 5 categories and 86.7% from 9 subcategories.	79% of the decline is concentrated among 4 manufacturers.	90% decline came from North America. Within it, 94.1% decline came from the US.
Prioritize customer behavior, competition, macroeconomic factors, inventory, marketing, & product mix.	Investigate what changed during these months and identify the factors contributing to the decline.	Prioritize affected products for targeted recovery actions.	Review supplier performance across pricing, quality, availability, and reliability.	Prioritize North America, particularly the US for recovery

FINANCIAL IMPACT

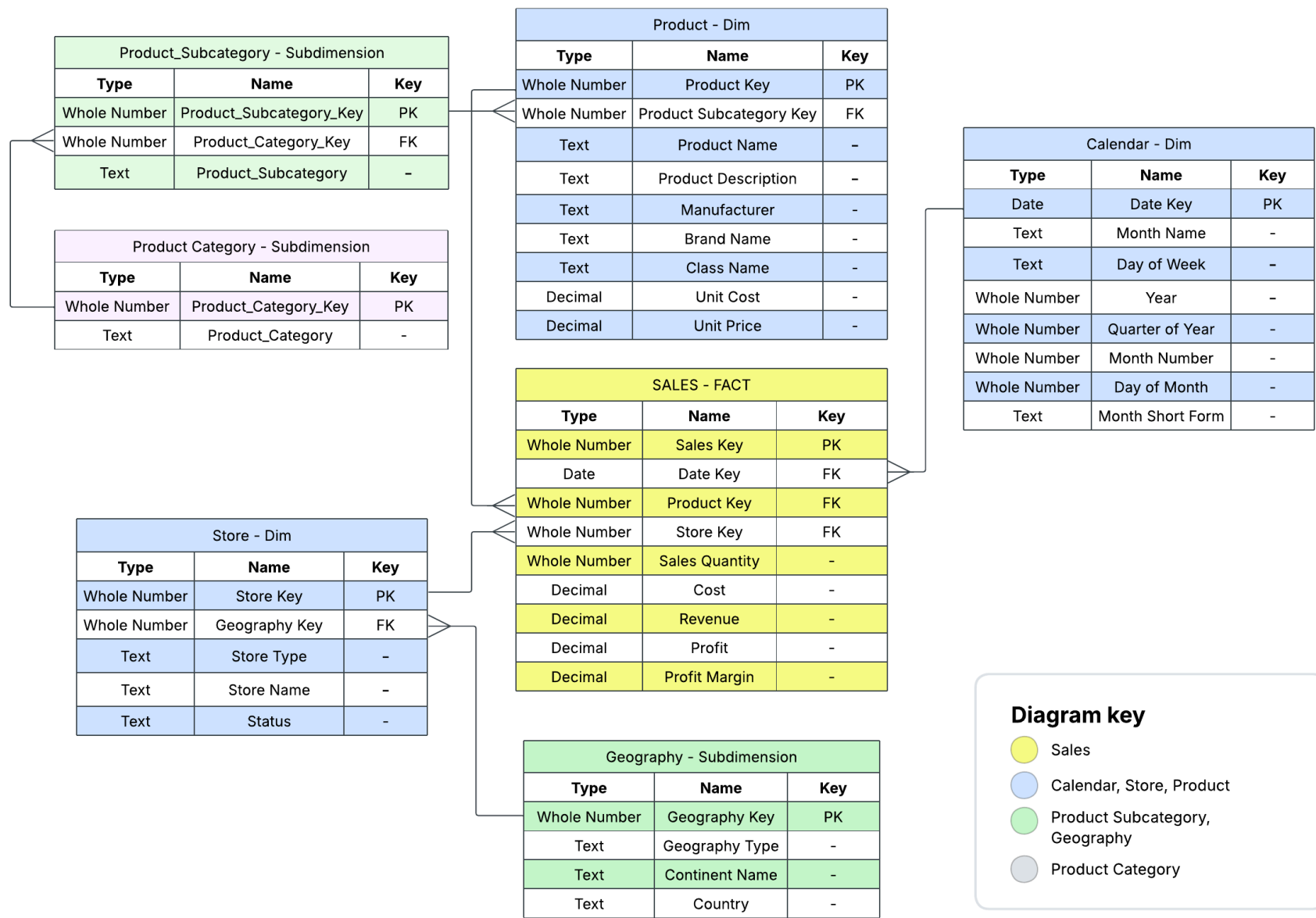
Gross Profit	Revenue Recovery
43.7% gross profit decline occurred alongside the 43.1% revenue decline.	Prioritize revenue recovery as the primary path to restoring gross profit.
Cost & Margin	Cost & Margin
42.4% cost decline did not offset the revenue loss, while profit margin declined 42.6%.	Maintain cost controls and monitor margin recovery alongside revenue recovery
All Dimensions	All Dimensions
All product categories, classes, manufacturers, continents, and countries contributed to the profit decline.	Focus recovery efforts on identified revenue drivers rather than broad cost-cutting measures.

RECOVERY ANALYSIS

Recovery Performance	Recovery Action
79.9% revenue decline, 43.4% quantity decline, and 4.4% ASP decline during the recovery period.	Apply root-cause actions for quantity decline & review pricing opportunities
Product & Supplier	Product & Supplier
35.4K units recovered across 4 categories & related subcategories. Top 5 manufacturers contributed to the recovery.	Identify recovery drivers and strengthen partnerships with high-performing manufacturers.
Geography	Geography
Asia was the top recovering continent, led by Pakistan, Armenia, China, and Japan.	Analyze recovery drivers in these markets and replicate successful approaches.



ENTITY RELATIONSHIP DIAGRAM



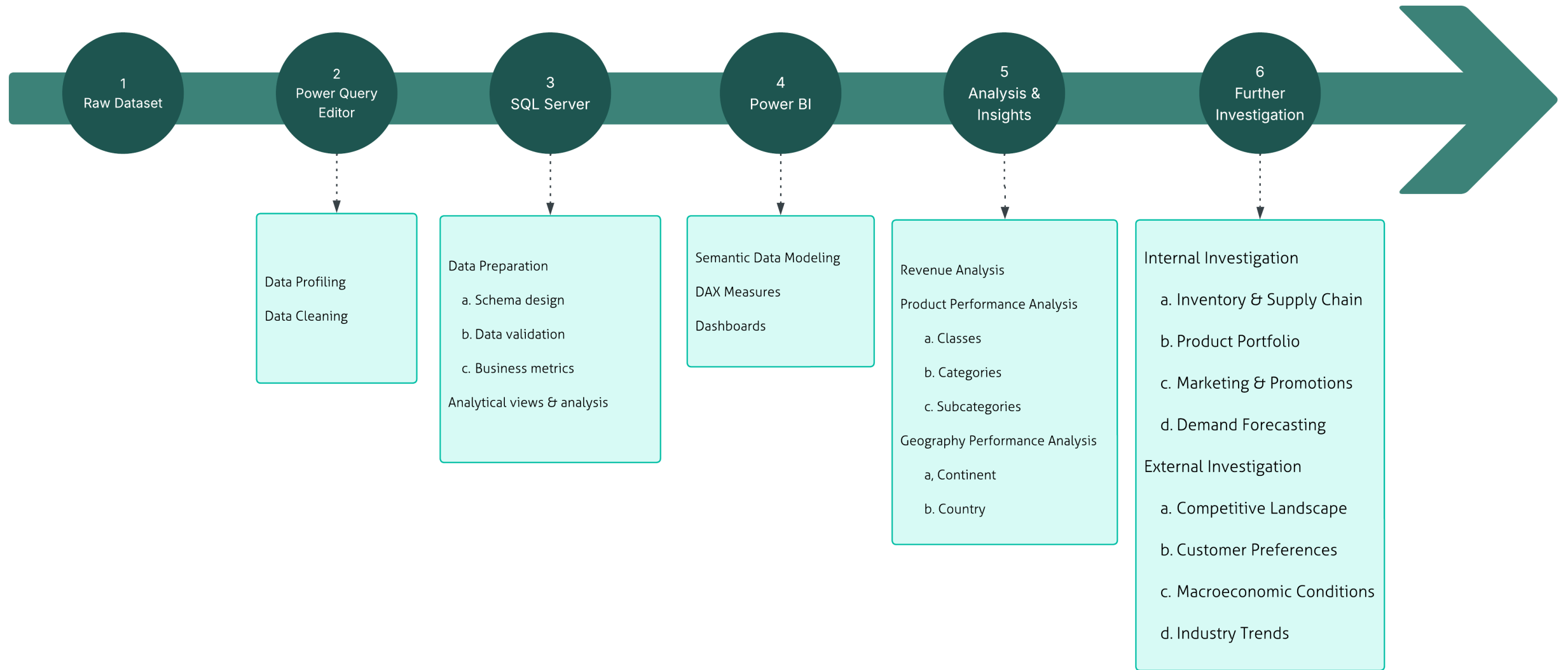


Table Name	Column Name	Description	Data Type	Key
Sales	Sales_Key	Unique identifier for each sales transaction	Whole Number	PK
Sales	Date_Key	Links each sales transaction to the Calendar table	Date	FK
Sales	Store_Key	Links each sales transaction to the Store table	Whole Number	FK
Sales	Product_Key	Links each sales transaction to the Product table	Whole Number	FK
Sales	Sales_Quantity	Number of units sold in each sales transaction	Whole Number	-
Sales	Revenue	Total Revenue generated from the sales transaction	Decimal Number	-
Sales	Profit	Total Profit earned from the sales transaction	Decimal Number	-
Sales	Profit_Margin	Profit expressed as a percentage of revenue for a sales transaction	Decimal Number (%)	-
Product	Product_Key	Unique identified for each product	Whole Number	PK
Product	Product_Subcategory_Key	Links each product to its corresponding product subcategory	Whole Number	FK
Product	Product_Name	Name of the Product	Text	-
Product	Product_Description	Detailed description of the product	Text	-
Product	Manufacturer	Name of the product manufacturer	Text	-
Product	Brand_Name	Brand under which the product is sold	Text	-
Product	Class_Name	Product classification	Text	-
Product	Unit_Cost	Cost incurred per unit of the product	Decimal Number	-
Product	Unit_Price	Selling Price per unit of the product	Decimal Number	-
Product_Category	Product_Category_Key	Unique identifier for each product category.	Whole Number	PK
Product_Category	Product_Category	Name of the product category	Text	-
Product_Subcategory	Product_Subcategory_Key	Unique identifier for each product subcategory.	Whole Number	PK
Product_Subcategory	Product_Category_Key	Name of the product subcategory	Whole Number	FK
Product_Subcategory	Product_Subcategory	Name of the product subcategory	Text	-
Stores	Store_Key	Unique identifier for each store.	Whole Number	PK
Stores	Geography_Key	Links each store to its corresponding geographic location.	Whole Number	FK
Stores	Store_Type	Type or format of the store	Text	-
Stores	Store_Name	Name of the store.	Text	-
Stores	Status	Operational status of the store (e.g., On or Off).	Text	-
Geography	Geography_Key	Unique identifier for each geographic location.	Whole Number	PK
Geography	Geography_Type	Classification of the geographic level (e.g., City, Country, Continent).	Text	-
Geography	Continent_Name	Name of the continent where the location is situated.	Text	-
Geography	Country	Name of the country associated with the geographic location.	Text	-
Calendar	Date_Key	Unique date identifier used to link transactions to the calendar.	Date	
Calendar	Month_Name	Full name of the month (e.g., April, May).	Text	
Calendar	Day_Of_Week_Name	Name of the day of the week (e.g., Monday, Tuesday).	Text	
Calendar	Year	Calendar year of the date.	Whole Number	
Calendar	Quarter_Of_Year	Quarter of the year (Q1-Q4).	Whole Number	
Calendar	Month_Of_Year	Numeric representation of the month (1-12).	Whole Number	
Calendar	Day_Of_Month	Day of the month (1-31).	Whole Number	
Calendar	Month_Short	3 letter abbreviation of the month (e.g., Jan, Feb, Mar). Used for concise reporting & visualizations.	Text	